

Title

A Digital Platform Targeting Symptom Burden in Inflammatory Bowel Disease: Study Protocol for Evaluating Clinical Effectiveness, Cost-Effectiveness, and Implementation

Abstract

Patients with inflammatory bowel disease frequently experience persistent pain, fatigue, and bowel dysfunction despite medical optimization. Digital self-management platforms offer potential for symptom-focused support, yet rigorous effectiveness and economic evaluation remains limited. This protocol describes a prospective parallel-group randomized trial comparing digital-facilitated self-management against standard care in approximately 780 adults with inflammatory bowel disease. The intervention comprises interactive modules addressing pain management, energy conservation, and continence strategies, delivered via secure platform with periodic facilitator contact. Participants will be randomized 1:1 to intervention or care-as-usual. Primary clinical outcomes include disease-specific quality-of-life metrics and global symptom improvement at 12 months. Secondary outcomes include healthcare utilization, out-of-pocket costs, work productivity, and quality-adjusted life years. Economic analysis will assess incremental cost-effectiveness ratios from both healthcare system and societal perspectives. Baseline characteristics, disease phenotype, and treatment intensity will be examined as potential effect modifiers.